BlueStock Mutual Fund Analytics Capstone Project

Prepared By

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1. Introduction

Mutual funds have become one of the most preferred investment options for individuals because they offer professional fund management and diversification. However, with hundreds of mutual fund schemes available in the market, comparing funds and identifying strong performers can be challenging.

The objective of this project is to analyze mutual fund data and evaluate fund performance using both return-based and risk-based metrics. The project focuses on understanding how different funds have performed over time, how much risk they carry, and how they compare against benchmark indices such as NIFTY 50 and NIFTY 100.

This capstone project combines data engineering, data analysis, financial analytics, database management, and reporting to build a complete mutual fund analytics solution.

2. Project Objective

The main objective of this project is to create a data-driven framework for analyzing and ranking mutual funds.

The project aims to:

- Collect and organize mutual fund datasets from multiple sources.
- Clean and validate data before analysis.
- Store and manage data efficiently using SQLite.
- Analyze historical NAV trends.

- Calculate important financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Compare mutual fund performance with benchmark indices.
- Create a composite scorecard for ranking funds.
- Generate reports and visualizations for better decision-making.

The final outcome is a system that helps investors understand which funds have delivered better returns while maintaining acceptable levels of risk.

3. Dataset Overview

The project uses multiple datasets related to the Indian mutual fund industry.

Fund Master Data

Contains information about mutual fund schemes and their AMFI codes.

Historical NAV Data

Contains daily NAV values for 40 mutual fund schemes and forms the foundation of all return calculations.

AUM Data

Provides Assets Under Management information for different fund houses, helping understand market share and growth trends.

SIP Inflow Data

Contains monthly SIP contribution data, allowing analysis of investor participation and investment behavior.

Benchmark Data

Contains historical values of benchmark indices such as NIFTY 50 and NIFTY 100 used for performance comparison.

Portfolio Holdings Data

Contains information about securities held by mutual fund schemes.

Overall, the project analyzed approximately **46,000 historical NAV records across 40 mutual fund schemes**.

4. Project Workflow

The project follows a structured workflow from raw data collection to final reporting.

Step 1: Data Ingestion

Raw datasets were collected and loaded into the project using Python.

Step 2: Data Cleaning

Data quality checks were performed to handle missing values, duplicate records, and formatting inconsistencies.

Step 3: Database Storage

The cleaned datasets were stored in SQLite for easier querying and management.

Step 4: Exploratory Data Analysis

Visualizations and summaries were created to understand trends in NAV, AUM, SIP inflows, and benchmark indices.

Step 5: Performance Analytics

Financial metrics were calculated to evaluate mutual fund performance and risk.

Step 6: Fund Ranking

A composite scorecard was developed to rank funds based on multiple performance indicators.

Step 7: Reporting

Insights, visualizations, and rankings were documented in reports and dashboards.

5. Data Cleaning and Preparation

Before performing any analysis, the data was carefully cleaned and validated.

The following steps were performed:

- Converted date columns into proper datetime format.
- Checked for missing values and handled them appropriately.
- Removed duplicate records.
- Standardized column names and formats.
- Verified AMFI code consistency across datasets.
- Validated NAV history records before calculating returns.

These steps ensured that the analysis was performed on accurate and reliable data.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the overall behavior of mutual fund data.

NAV Trend Analysis

Historical NAV values were analyzed to observe fund growth over time.

AUM Analysis

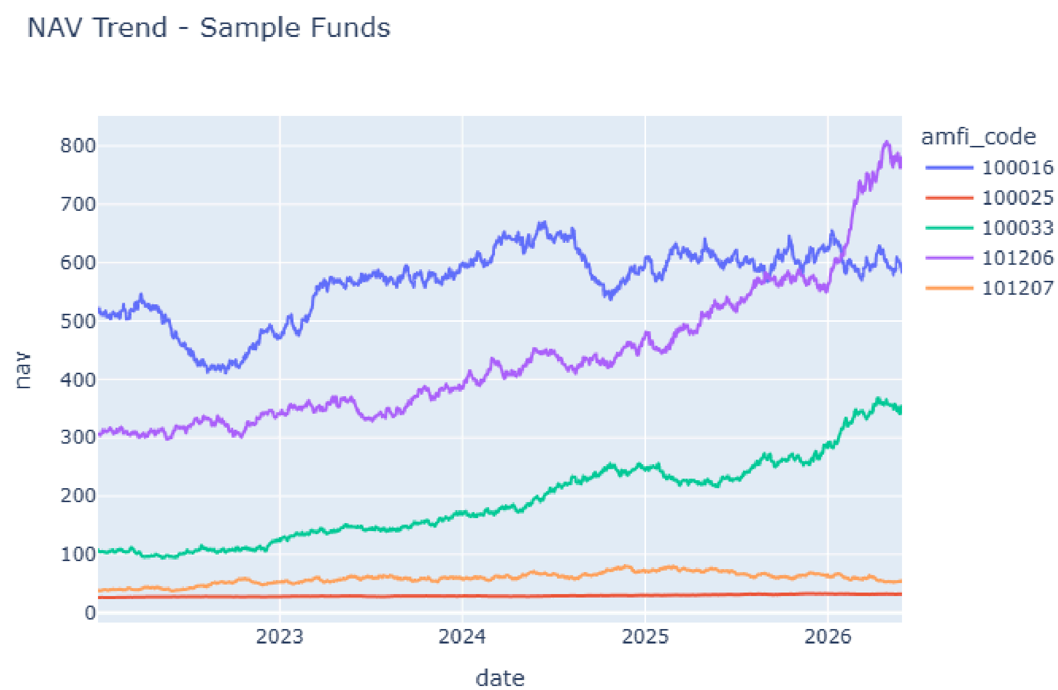
Assets Under Management data was used to identify the largest fund houses and understand market concentration.

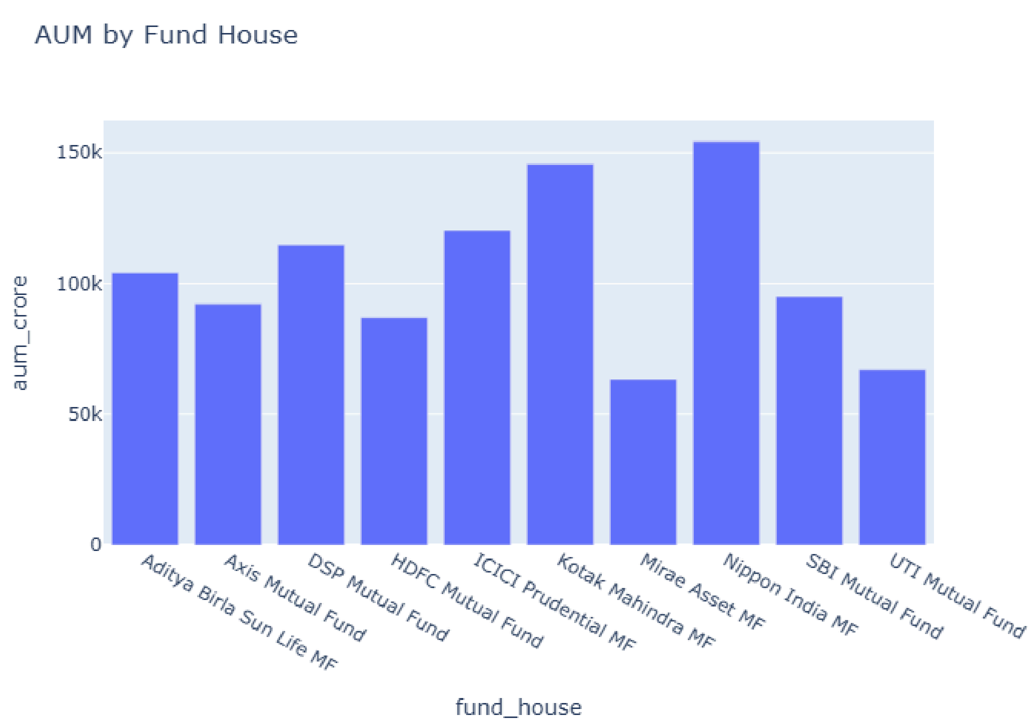
SIP Trend Analysis

Monthly SIP inflows were studied to understand investor participation and long-term investment trends.

Benchmark Analysis

Benchmark indices such as NIFTY 50 and NIFTY 100 were analyzed to understand overall market movement during the study period.





7. Performance Analytics

Several financial and risk-adjusted metrics were calculated to evaluate mutual fund performance.

Daily Returns

Daily returns were calculated using the formula:

Daily Return = (NAV_t / NAV_{t-1}) − 1

This metric measures short-term fund performance and serves as the foundation for further risk calculations.

CAGR (Compound Annual Growth Rate)

CAGR measures the annualized growth rate of an investment over a specified period.

Formula:

CAGR = (NAV End / NAV Start)^(1/n) − 1

The project calculated:

- 1-Year CAGR
- 3-Year CAGR

Several funds generated strong annualized returns over the analyzed period.

Sharpe Ratio

The Sharpe Ratio measures risk-adjusted returns by comparing excess returns with total volatility.

Formula:

Sharpe Ratio = $(R_p - R_f) / \sigma \times \sqrt{252}$

Where:

- Rp = Portfolio Return
- Rf = Risk-Free Rate
- σ = Standard Deviation of Returns

Funds with higher Sharpe Ratios demonstrated better risk-adjusted performance.

Sortino Ratio

The Sortino Ratio is similar to the Sharpe Ratio but considers only downside volatility.

This provides a more focused measure of risk by penalizing only negative return fluctuations.

Funds with high Sortino Ratios showed strong returns while limiting downside risk.

Alpha and Beta

Alpha and Beta were calculated using linear regression against the NIFTY 100 benchmark.

Alpha

Measures excess return generated beyond the benchmark.

Beta

Measures sensitivity of a fund to benchmark movements.

These metrics help evaluate fund manager effectiveness and market exposure.

Maximum Drawdown

Maximum Drawdown measures the largest decline from a peak value.

It helps investors understand the worst historical loss experienced by a fund.

Some funds experienced significant drawdowns during market corrections, highlighting the importance of risk management.

8. Fund Scorecard

A composite scorecard was developed to rank funds based on multiple performance indicators.

The score was calculated using:

- 40% CAGR Rank
- 30% Sharpe Ratio Rank
- 20% Alpha Rank
- 10% Maximum Drawdown Rank

Scores were normalized to a scale of 0–100.

This approach provides a balanced evaluation by considering returns, risk-adjusted performance, and downside protection.

9. Benchmark Comparison

The top-performing funds were compared against:

- NIFTY 50
- NIFTY 100

Tracking Error was also calculated to measure how closely funds followed benchmark performance.

The comparison helped identify funds that consistently outperformed the broader market.

10. Key Outcomes

The project successfully achieved its objectives and produced meaningful insights.

Major outcomes include:

- Processed and analyzed over 46,000 NAV records.

- Evaluated 40 mutual fund schemes.
- Calculated return-based and risk-based performance metrics.
- Built a composite fund ranking system.
- Generated benchmark comparison analytics.
- Created a structured workflow covering data ingestion, cleaning, analysis, and reporting.

The project demonstrates how analytics can support investment evaluation and decision-making.

11. Challenges Faced

Several challenges were encountered during the project.

Data Cleaning Challenges

The datasets required extensive preprocessing before analysis could begin.

Benchmark Alignment

Ensuring that benchmark index data and mutual fund NAV data aligned correctly by date was a key challenge.

Financial Metric Implementation

Implementing financial metrics accurately required both financial understanding and technical implementation.

Debugging and Validation

Multiple calculations had to be validated to ensure realistic and accurate results.

These challenges helped strengthen problem-solving and analytical skills.

12. Future Enhancements

Several improvements can be added in future versions of the project.

Real-Time NAV Integration

Integrate live NAV data using APIs for real-time analytics.

Interactive Dashboard

Develop a complete Power BI dashboard with advanced filtering and drill-down features.

Fund Recommendation System

Use Machine Learning techniques to recommend suitable funds based on investor preferences.

Portfolio Optimization

Apply Modern Portfolio Theory to suggest optimized investment portfolios.

Automated Reporting

Generate periodic reports automatically without manual intervention.

Cloud Deployment

Deploy the solution on cloud platforms for wider accessibility.

13. Conclusion

This capstone project successfully demonstrates an end-to-end mutual fund analytics workflow.

The project combines data engineering, database management, financial analytics, visualization, and reporting to evaluate mutual fund performance comprehensively.

By calculating metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown, the project provides a detailed understanding of both returns and risk. The composite fund scorecard further enables objective fund comparison and ranking.

Overall, this project highlights the practical application of Python, SQL, SQLite, GitHub, and data analytics techniques in the financial domain and serves as a strong foundation for future enhancements such as real-time analytics and intelligent investment recommendation systems.